

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

First Semester of B.Tech. (CE/EE/IT) (EC/ME/CL)

December 2010

CE – 101 Fundamentals of Computing & Programming

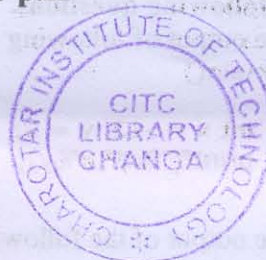
Date: 13/12/2010, Monday

Time: 10:30 a.m to 1:30 p.m

Maximum Marks:70

Instructions:

1. Attempt all questions.
2. Answers both sections in separate answer books.
3. Figure to the right indicates full marks.
4. Assume suitable data, if necessary and mention it.



FOR REFERENCE

SECTION – I

Q.1(a) Answer the following questions.

05

1. To read an integer value in variable named number is use of following statement correct? Answer with reason. `scanf("%d",number);`
2. The Operator _____ can't be used with real operands and the _____ operator returns the number of bytes the operand occupies.
3. Name the function from string.h library which helps to compare two strings.
4. State true or false: Every switch statement must have default case included.
5. What is the return value of `printf()` function?

Q.1(b) What is the correct output of the following code?

06

(1) <code>void main ()</code> <code>{ int y = 9, x = 4;</code> <code>if(x = 3)</code> <code>printf ("%d",y);</code> <code>else</code> <code>printf ("%d",x);</code> <code>}</code>	(2) <code>void main ()</code> <code>{</code> <code>int x=10,y=11,z=12;</code> <code>printf("%d");</code> <code>}</code>	(3) <code>#define square(X) X*X</code> <code>void main ()</code> <code>{ int Y=16,X=4,ans;</code> <code>ans=Y/square(X);</code> <code>printf("%d",ans); }</code>
---	--	---

Q.2 Write programs in C.

12

1. Write a program to find the reverse of a given number.
2. Write a program to implement user interactive calculator. The program allows four operation +, -, *, /. For / operation, also print an error if division is not possible.

OR

Q.2 Write programs in C.

12

1. Write a program to find the complement of input number of any length. (ex. The complement of 123 is $999 - 123 = 876$)
2. Write a program to convert a decimal number into octal number.

Q.3 Answer the following questions.

12

1. Explain nested if-else and switch statement with an example.
2. Which are the methods of passing a parameter to function? Explain them with an example.
3. Explain bitwise operator with an example.

OR

Q.3 Answer the following questions.

12

1. Explain two-dimensional array for integer with an example.
2. State true or false: It is possible to return multiple values by a function. Answer with reason and give suitable example.
3. Explain Basic Structure of C Program.

SECTION – II

Q.4 Answer the following questions.

11

1. What is the output of following program?

02

```
void main()
{
    int x = -12, y = 5;
    printf("%d", x%y);
}
```

2. What is the output of the following program?

02

```
void main()
{
    printf("here we go");
    return;
    printf("yes I'll be executed");
}
```

3. What is the scope and lifetime of static, automatic and external variables.
4. What is the difference between variable pointer and constant pointer?
5. Write a recursive function to find the factorial of a given number.

03

02

02

Q.5 Write programs in C.

12

1. Write a program to check whether the given string is palindrome or not without using string.h library.
2. Write a user-defined function which multiplies two square matrices and also test this function.

OR

Q.5 Write programs in C.

12

1. Write a program that takes following information about each student as an input.
Student Name, Marks of 1st test, Marks of 2nd test, Marks of 3rd test
The program allows entry of maximum 30 students. After entry of data is over, the program prints total marks of each student along with input data.
2. Write a user-defined function which will take two-dimensional array of words and arrange them in dictionary order and also test this function.

Q.6 Answer the following questions.

12

1. What is dynamic memory allocation? Discuss its advantages.
2. Explain strlen(), strcat(), strcpy() functions.
3. Explain pointers to array with an example.
4. Write a program to copy one file to another file.

OR

Q.6 Answer the following questions.

12

1. Which are the modes of file? Explain fseek() function related to file.
2. Differentiate between structure and union.
3. Explain pointer to pointer with an example.
4. Write a program using pointers to read an array of integers and print its elements in reverse order.